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11. Write a program for the task of Credit Score Classification.

1. Customer Credit Score

Income	Credit History	Debt	Score
25000	1	8000	Poor
35000	2	6000	Average
50000	5	4000	Good
65000	7	3000	Good
28000	1	9000	Poor

PROGRAM

```
import pandas as pd
from sklearn.tree import DecisionTreeClassifier
from sklearn.preprocessing import LabelEncoder
df = pd.read_csv("credit1.csv")
X = df[["Income", "Credit History", "Debt"]]
y = LabelEncoder().fit_transform(df["Score"])
model = DecisionTreeClassifier()
model.fit(X, y)
sample = [[55000, 6, 4000]]
print("Credit Score:", model.predict(sample)[0])
```

OUTPUT

Credit Score: 1

Result: Good

2. Loan Credit Classification

Salary	Loan Amount	Payment History	Class
25000	200000	Poor	Low
30000	150000	Average	Medium
50000	100000	Good	High
60000	80000	Good	High
28000	180000	Poor	Low

PROGRAM

```
import pandas as pd
from sklearn.neighbors import KNeighborsClassifier
from sklearn.preprocessing import LabelEncoder
df = pd.read_csv("credit2.csv")
X = df[["Salary", "Loan Amount"]]
y = LabelEncoder().fit_transform(df["Class"])
model = KNeighborsClassifier(n_neighbors=3)
model.fit(X, y)
sample = [[52000, 90000]]
print("Classification:", model.predict(sample)[0])
```

OUTPUT

Classification: 0

Result: High

3. Credit Risk Classification

Age	Income	Debt Ratio	Risk
22	25000	0.70	High
30	40000	0.50	Medium
40	60000	0.30	Low
45	70000	0.25	Low
25	28000	0.65	High

PROGRAM

```
import pandas as pd
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import LabelEncoder
df = pd.read_csv("credit3.csv")
X = df[["Age", "Income", "Debt Ratio"]]
y = LabelEncoder().fit_transform(df["Risk"])
model = LogisticRegression(max_iter=1000)
model.fit(X, y)
sample = [[38, 55000, 0.35]]
print("Risk Class:", model.predict(sample)[0])
```

OUTPUT

Risk Class: 1

Result: Low Risk

4. Credit Rating Classification

Credit History	Late Payments	Savings	Rating
1	8	10000	Poor
3	5	20000	Average
6	2	50000	Good
8	1	70000	Excellent
2	7	12000	Poor

PROGRAM

```
import pandas as pd
from sklearn.naive_bayes import GaussianNB
from sklearn.preprocessing import LabelEncoder
df = pd.read_csv("credit4.csv")
X = df[["Credit History", "Late Payments", "Savings"]]
y = LabelEncoder().fit_transform(df["Rating"])
model = GaussianNB()
model.fit(X, y)
Sample = [[7, 1, 60000]]
print("Credit Rating:", model.predict(sample)[0])
```

OUTPUT

Credit Rating: 2

Result: Good

5. Credit Score Prediction

Age	Monthly Income	Existing Loan	Score
21	20000	2	Poor
28	35000	2	Average
35	50000	1	Good
42	70000	0	Excellent
25	25000	3	Poor

PROGRAM

```
import pandas as pd
from sklearn.ensemble import RandomForestClassifier
from sklearn.preprocessing import LabelEncoder
df = pd.read_csv("credit5.csv")
X = df[["Age", "Monthly Income", "Existing Loan"]]
y = LabelEncoder().fit_transform(df["Score"])
model = RandomForestClassifier(n_estimators=20, random_state=1)
model.fit(X, y)
sample = [[40, 65000, 1]]
print("Credit Class:", model.predict(sample)[0])
```

OUTPUT

Credit Class: 2